=== CONFIGURACION GNB (1779968058) ===

```yaml
# gnb.yaml
# srsRAN Project gNB configuration file
#
# This example configuration outlines how to configure the srsRAN Project gNB to create a single NR cell
# transmitting in band n78, with 20 MHz bandwidth and 15 kHz sub-carrier-spacing.
# ZMQ is configured as the RF frontend for virtual radio communication.

cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.10
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1

ru_sdr:
  device_driver: zmq
  device_args: "tx_port=tcp://10.53.1.20:2000,rx_port=tcp://0.0.0.0:2001,base_srate=30.72e6"
  srate: 30.72
  tx_gain: 80
  rx_gain: 40

cell_cfg:
  dl_arfcn: 630000
  band: 78
  channel_bandwidth_MHz: 20
  common_scs: 15
  ssb_scs: 15
  plmn: "00101"
  tac: 7
  nof_prb: 100
  pdcch:
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
    common:
      ss0_index: 0
      coreset0_index: 12
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64

log:
  filename: /tmp/gnb.log
  all_level: info

pcap:
  mac_enable: enable
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: enable
  ngap_filename: /tmp/gnb_ngap.pcap
```

=== CONFIGURACION UE (1779968058) ===

```
# ue.conf
# srsRAN Project UE configuration file
#
# This example configuration outlines how to configure the srsRAN Project UE to connect to a NR cell
# transmitting in band n78, with 20 MHz bandwidth and 15 kHz sub-carrier-spacing.
# ZMQ is configured as the RF frontend for virtual radio communication.

[rf]
freq_offset = 0
tx_gain = 10
rx_gain = 5
srate = 30.72e6
nof_antennas = 1
device_name = zmq
device_args = "tx_port=tcp://10.53.1.10:2001,rx_port=tcp://0.0.0.0:2000,base_srate=30.72e6"
time_adv_nsamples = auto

[rat.nr]
bands = 78
nof_carriers = 1
max_nof_prb = 100
nof_prb = 100
dl_nr_arfcn = 630000
ssb_nr_arfcn = 630000

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc  = 63BFA50EE6523365FF14C1F45F88737D
k    = 00112233445566778899aabbccddeeff
imsi = 001010123456789
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
# ip_addr is intentionally omitted for dynamic assignment by the Core

[gui]
enable = true
```

=== DOCKER COMPOSE (1779968058) ===

```yaml
# docker-compose.yml
# Docker Compose file for Open5GS Core, srsRAN gNB, and srsRAN UE
#
# This configuration sets up a 5G network simulation with virtual radio (ZMQ)
# using Open5GS for the Core Network and srsRAN for the gNB and UE.

version: '3.8'

services:
  mongodb:
    image: mongo:4.4
    container_name: mongodb
    restart: always
    networks:
      ran:
        ipv4_address: 10.53.1.100 # MongoDB can be anywhere, but in the ran network

  open5gs-amf:
    image: open5gs/open5gs
    container_name: open5gs-amf
    command: ["open5gs-amfd"]
    volumes:
      - ./config/open5gs-amf.yaml:/usr/local/etc/open5gs/amf.yaml
    networks:
      ran:
        ipv4_address: 10.53.1.2
    depends_on:
      - mongodb
    restart: always

  open5gs-upf:
    image: open5gs/open5gs
    container_name: open5gs-upf
    command: ["open5gs-upfd"]
    volumes:
      - ./config/open5gs-upf.yaml:/usr/local/etc/open5gs/upf.yaml
    networks:
      ran:
        ipv4_address: 10.53.1.3
    depends_on:
      - mongodb
    restart: always
    privileged: true # Required for UPF to create tun device

  open5gs-webui:
    image: open5gs/webui
    container_name: open5gs-webui
    ports:
      - "3000:3000"
    networks:
      ran:
        ipv4_address: 10.53.1.4
    depends_on:
      - mongodb
    restart: always

  gnb:
    image: srsran/enb
    container_name: srsran-gnb
    privileged: true
    cap_add:
      - SYS_NICE
      - CAP_SYS_PTRACE
    volumes:
```

```
      - /tmp:/tmp
      - ./gnb.yaml:/etc/srsran/gnb.yaml
    command: srsgnb /etc/srsran/gnb.yaml
    networks:
      ran:
        ipv4_address: 10.53.1.10
    depends_on:
      - open5gs-amf
    restart: always

  ue:
    image: srsran/ue
    container_name: srsran-ue
    privileged: true
    cap_add:
      - SYS_NICE
      - CAP_SYS_PTRACE
    volumes:
      - /tmp:/tmp
      - ./ue.conf:/etc/srsran/ue.conf
    command: srsue /etc/srsran/ue.conf
    networks:
      ran:
        ipv4_address: 10.53.1.20
    depends_on:
      - gnb
    restart: always

networks:
  ran:
    name: ran
    driver: bridge
    ipam:
      config:
        - subnet: 10.53.1.0/24

# Configuration files for Open5GS (to be placed in a 'config' directory)
# config/open5gs-amf.yaml
# logger:
#   level: info
#   report:
#     - all
# global:
#   mcc: 001
#   mnc: 01
#   nssai:
#     sst: 1
#     sd: 0x010203
#   served_guami:
#     - plmn_id:
#         mcc: 001
#         mnc: 01
#       amf_region_id: 0x01
#       amf_set_id: 0x01
#       amf_pointer: 0x01
#   tmsi:
#     amf_set_id: 0x01
#     amf_pointer: 0x01
#   security:
#     integrity_order:
#       - NIA2
#       - NIA0
#     ciphering_order:
#       - NEA0
#       - NEA2
# amf:
#   sbi:
#     server:
```

```
#       - addr: 10.53.1.2
#         port: 80
#   ngap:
#     server:
#       - addr: 10.53.1.2
#   metrics:
#     server:
#       - addr: 10.53.1.2
#         port: 9090

# config/open5gs-upf.yaml
# logger:
#   level: info
#   report:
#     - all
# upf:
#   sbi:
#     server:
#       - addr: 10.53.1.3
#         port: 80
#   pfcp:
#     server:
#       - addr: 10.53.1.3
#   gtpu:
#     server:
#       - addr: 10.53.1.3
#   metrics:
#     server:
#       - addr: 10.53.1.3
#         port: 9090
```